1、

Pr(至少一次错误) $\le t\epsilon$ $\quad Pr$(一直没错) $\ge 1-t\epsilon$

$$p_{\pi_A}(s_t) = (1-\delta_t)\,p_{\pi^*}(s_t) + \delta_t\, p_{mistake}(s_t) \qquad \delta_t \le t\epsilon$$

$$p_{\pi_A}(s_t) - p_{\pi^*}(s_t) = \delta_t\left(p_{mistake}(s_t) - p_{\pi^*}(s_t)\right)$$

$$\sum_{s_t}\left|p_{\pi_A}(s_t) - p_{\pi^*}(s_t)\right| = \delta_t \sum_{s_t}\left|p_{mistake}(s_t) - p_{\pi^*}(s_t)\right|$$

$$\le 2\delta_t \quad \le 2t\epsilon \quad \le 2T\epsilon$$

2、 $|f(s)| \le M$ $\quad \left|E_p f(s) - E_q f(s)\right| \le M \sum_s |p(s) - q(s)|$

$$\left|E_{p_{\pi^*}(s_t)} r(s_t) - E_{p_{\pi_A}(s_t)} r(s_t)\right| \le R_{max} \sum_s \left|p_{\pi^*}(s_t) - p_{\pi_A}(s_t)\right|$$

由1得 $\le 2R_{max}\,t\epsilon$

(a). $J(\pi) = E_{p_\pi(s_T)}\, r(s_T)$

$$J(\pi^*) - J(\pi_A) = E_{p_{\pi^*}(s_T)}\, r(s_T) - E_{p_{\pi_A}(s_T)}\, r(s_T) \le 2R_{max}T\varepsilon$$

$$\therefore J(\pi^*) - J(\pi_A) = O(T\varepsilon)$$

(b).

$$J(\pi^*) - J(\pi_A) = \sum_{t=1}^{T} E_{p_{\pi^*}(s_t)}\, r(s_t) - E_{p_{\pi_A}(s_t)}\, r(s_t)$$

$$\le \sum_{t=1}^{T} 2R_{max}\, t\varepsilon = 2R_{max}\varepsilon \frac{(1+T)T}{2}$$

$$\therefore J(\pi^*) - J(\pi_A) = O(T^2\varepsilon)$$